DATA ITEM DESCRIPTION

Title: CYBERSECURITY SUPPLY CHAIN RISK MANAGEMENT HARDWARE BILL OF MATERIALS REQUIREMENTS

Number: DI-SESS-82434
AMSC Number: N10453
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20240119
Limitation: N/A
GIDEP Applicable: N/A
Project Number: SESS-2024-001

Use/relationship: This Data Item Description (DID) is designated to ensure that suppliers and vendors are actively engaged in maintaining and improving Cybersecurity Supply Chain Risk Management (C-SCRM) practices. The DID mandates the creation and provision of a comprehensive Hardware Bill of Materials (HBOM). It provides a standardized and systematic methodology for cataloging hardware components within embedded and connected devices to facilitate better understanding and management of potential cybersecurity risks.

a. The HBOM provided must be a representation of the production release manufacturing bill of materials, or "as-built" BOM representing the components that were actually used in the production of the product. These should include (where applicable)

(1) Configurable BOM (CBOM) – These BOMs describe multiple options in the manufacturing process driven by selection of options for unique models in a product family. The supplier must provide all iterations of the BOM that are relevant for the "as-built" models procured by the customer.

(2) BOM Variant – These BOMs describe multiple options in the manufacturing process used for the same model of a product. For instance, one server may use network adapter cards from multiple vendors with the same form, fit and function. If BOM variants are used for this purpose, the supplier must provide all of the relevant BOM variants for the "as-built" models procured by the customer.

(3) Manufacturing Deviations - Any exceptions to approved or "as-built" manufacturing that have been applied. These are temporary exceptions to documented manufacturing procedure or process that are put in place until a permanent change can be made. For instance, a component may have been specified with a tighter tolerance than actually needed (1% instead of 5%) for form, fit, and function. A deviation can be written to allow use of 5% parts for a specified period of time while a change order is written to replace the 1% parts.

b. Unless otherwise specified hardware BOM type should not include:

(1) Engineering BOM (EBOM) – Used in the design of the product, but may not be an accurate representation of the manufactured product. This would only be appropriate in the prototype phase of a project.

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c. The word “item” is used throughout the document to refer to hardware components of the system.

d. This DID contain the format, content, and intended use information for the data product resulting from the work task described in the solicitation.

e. This DID should be tailored in the Contract Data Requirements List (CDRL) to include the necessary field categories from Table 1.

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

1.1. Hardware Bill of Materials (HBOM) Framework for Supply Chain Risk Management, Hardware Bill of Materials (HBOM) Framework for Supply Chain Risk Management | CISA

2. Format: The C-SCRM HBOM shall be submitted in English and machine-readable format to ensure interoperability and automation support.

3. Content: The data provided in the HBOM must adhere to the standards and specifications set forth in this document. As stipulated by the contract, the data must encompass the information outlined in the Field Name and Description columns detailed in Table 1. Fields for which data is not available shall be left blank. Vendors are required to provide an HBOM breakdown as shown in Figure 1, demonstrating a hierarchical decomposition of the product.

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FIGURE 1

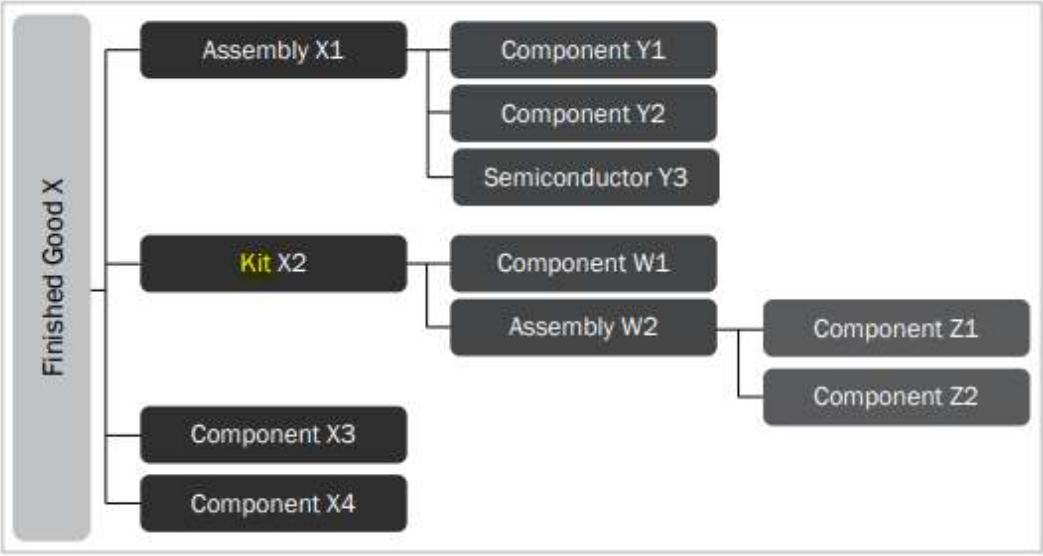


TABLE 1

Field Name	Required	Description
Manufacturer	Yes	Manufacturer of the component
Mfg. Cage Code	Yes	Cage Code of Manufacturer
Mfg. Part Number	Yes	Manufacturer part number for the component
Manufacturer 2	Yes	Alternate Manufacturer of the component
Mfg. 2Part Number	Yes	Alternate Manufacturer part number for the component
Manufacturer 3	Yes	Second Alternate Manufacturer of the component
Mfg. 3 Part Number	Yes	Second Alternate Manufacturer part number for the component
Quantity	Yes	Quantity of stated component
Function	Yes	Bluetooth, network, storage, microprocessor, connector, etc.
Reference Designator	Yes	Component Identifier on schematic and pcba
Location	Supported	Location on the board, related daughterboards
Component Type	Yes	SMD, thru-hole, etc.
Description	Yes	Manufacturer description
Serial Number	Supported	Unique identifier using serial number if available

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MTBF	Supported	Mean Time Between Failure (MTBF) refers to the average amount of time that a device or product functions before failing.
Lead Time	Supported	The time between the manufacturers' receipt of an order for the item and delivery.
Lifecycle Status	Supported	Production Status of the item from the manufacturer Can be ACTIVE, NRND (Not recommended for New Designs), Last Time Buy, Obsolete, etc.
SKU	Supported	Internal inventory reference if available
Lot number	Supported	Lot or batch identification for the component
Production timestamp	Supported	Production timestamp for the component
Manufacturing address	Supported	Manufacturing location
Programming location	Supported	Programming location if different from manufacturing loc
MAC Address	Supported	Hardware address for network interfaces
FCC ID	Supported	FCC identifier for radio components
Other	Supported	Use of other information such as QC name, or build instructions
EAN/UPC	Supported	Instantly-recognizable, EAN/UPC barcodes are printed on virtually every consumer product in the world.
Global Trade Item Number (GTIN)	Supported	Global Trade Item Number (GTIN) can be used by a company to uniquely identify all of its trade items. GS1 defines trade items as products or services that are priced, ordered or invoiced at any point in the supply chain.
Global Location Number (GLN)	Supported	Global Location Number (GLN) can be used by companies to identify their locations, giving them complete flexibility to identify any type or level of location required.
Global Individual Asset Identifier (GIAI)	Supported	The Global Individual Asset Identifier is one of the two GS1 Keys for asset identification. Companies can apply a GIAI on any asset to uniquely identify and manage that asset. This could be a computer, desk, vehicle, piece of transport equipment, or spare part, as just a few examples.
Global Model Number (GMN)	Supported	The Global Model Number enables users to uniquely identify the product model through the entire life cycle of the product: design - production – procurement – use – maintenance - disposal.

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EPC/RFID	Supported	The Electronic Product Code™ (EPC) is syntax for unique identifiers assigned to physical objects, unit loads, locations, or other identifiable entity playing a role in business operations.

End of DI-SESS-82434

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